DeskBeam Logbook

Entry #1: 9/15/2026

As a group, we created a ToDos list regarding the general scope of our project. We came together as a group to better understand the goals, timeline, approach, software, responsibilities, and schedule for our project. We also came up with the name “DeskBeam,” encompassing a bright autonomous study lamp suited for anyone and everyone.

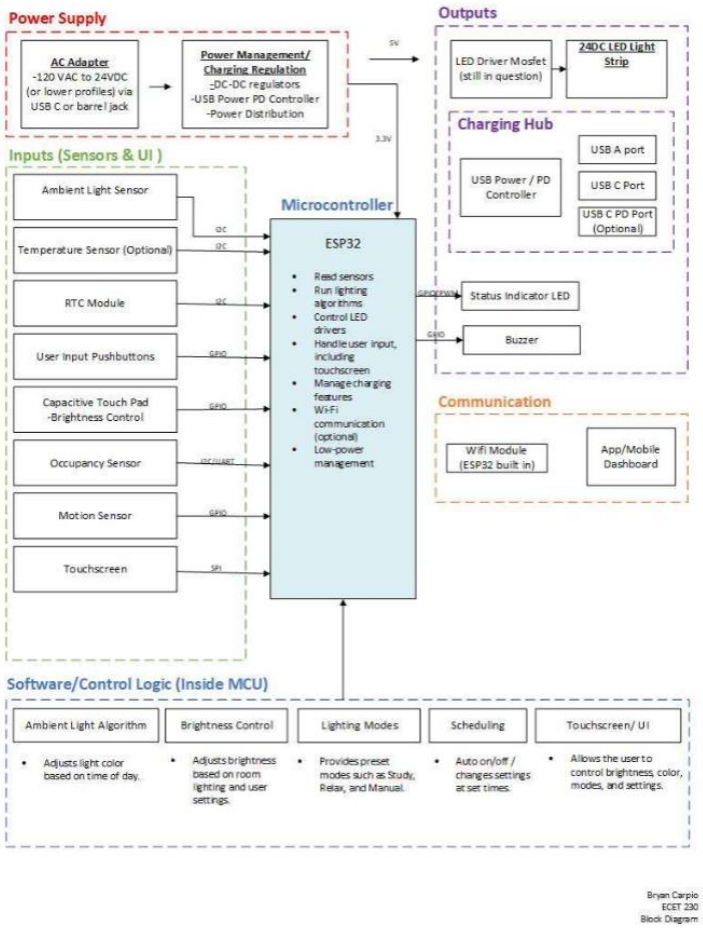
Entry #2: 9/15/2026

Given our understanding behind DeskBeam, we are now fully aware of what is needed for the project to function accordingly. Brief component selection was done. Rough draft of a block diagram was created. This was pivotal to understanding the architecture of our project. Further component selection will be done during the next couple of days. Research behind components must be done as well to ensure high quality and effective dev boards are chosen for the initial POC prototype.

Entry #3: 9/22/2026

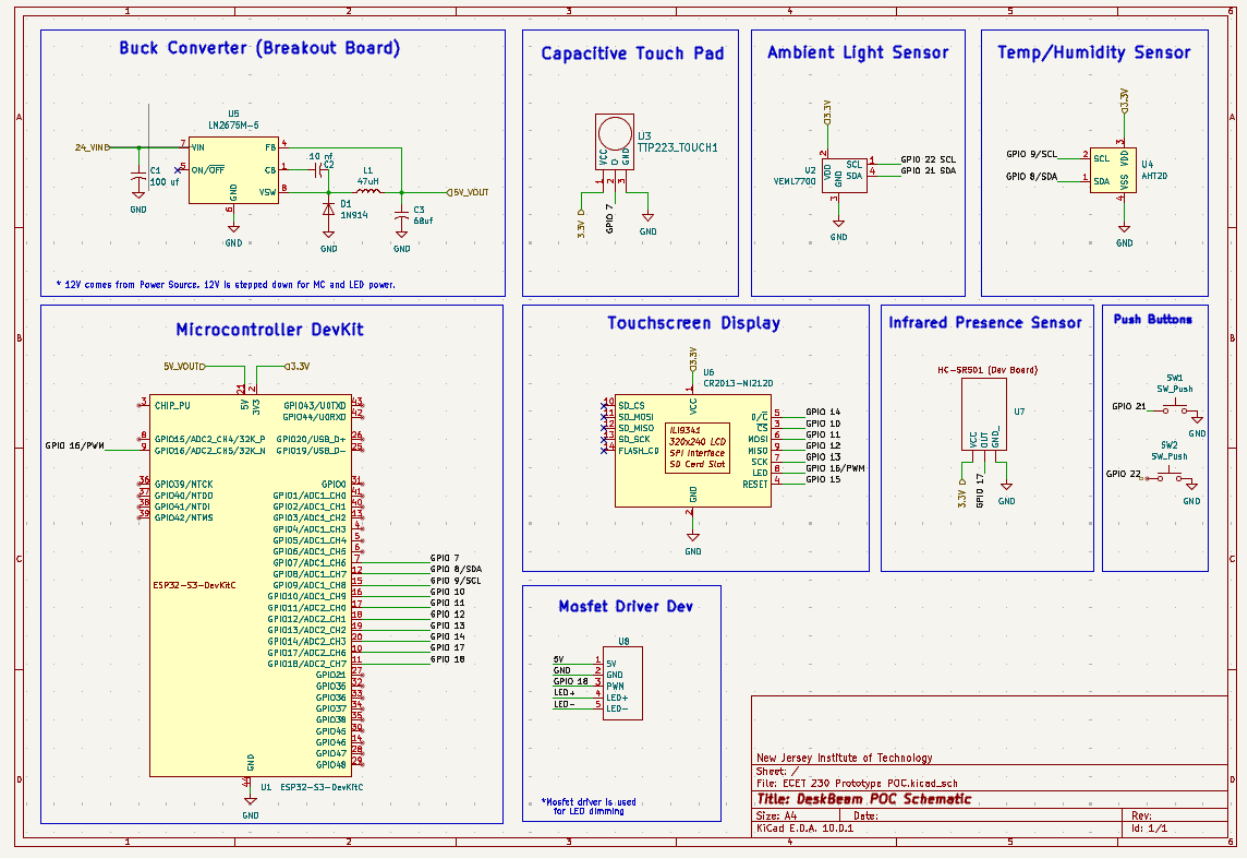
Finalized block diagram. Attached below. Component selection was completed. Most of the dev boards will be obtained from either Amazon or Digikey given time restraints.

Block Diagram (Visio):



Entry #4: 9/29/2026:

Today I started the schematic project on Kicad. Initially, we started with the hierarchal blocks as demonstrated and discussed in class, however, I felt that since our upcoming prototype would only be the POC, the most helpful and effective way to go about the schematic would be to create one of only the development boards being used. I felt this was best to fully create an overview of how everything for the POC was going to be wired up. I downloaded footprints and symbols from open source platforms such as GitHub. Overall, pretty content with the readability and overall design of the schematic. Attached below is a screenshot of the schematic.



Entry #5: 9/29/2026:

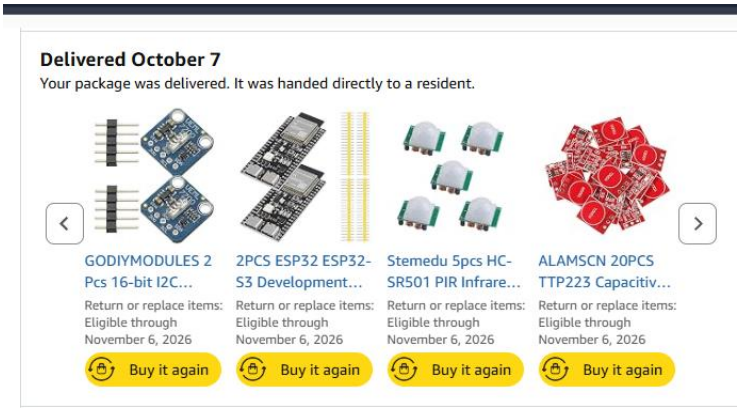
On top of the schematic being completed, we also created a prototype plan. We will go about the overall design of DeskBeam in this order: POC Prototype (Mauled), Looks-like prototype (LeBron), works like prototype (Big Ben), and finally, the engineering prototype (Lionel Andrés Messi Cuccittini).

Entry #6: 10/5/2026

All the components'/dev boards were ordered via Amazon. Given time constraint, we settled with amazon for all our parts. Multiple parts were ordered in case one came back defective. Datasheets were downloaded and put into a folder.

Entry #7: 10/7/2026

All components for the DeskBeam project were successfully delivered. I also took the liberty of testing most of the components individually to verify their functionality before beginning the integration process. All sensors tested worked as expected, with no noticeable issues or defects. These initial tests helped confirm that the components were operational and ready for use in the prototype.



Entry #8: 10/17/2026:

I began the POC prototyping phase. I started with the MCU and testing verification. This was the only component that I did not test. Somehow, the ESP32 S3 came defective, both. Given the time constraint, I did not have enough time to reorder them. So, I resorted to the Node MCU ESP32 S, which I had laying around from a previous project. Luckily, it worked with no issues. I began integrating each sensor one by one on the bread one, testing each one with its own respective program to ensure functionality. They all worked.

Entry #9: 10/18/2026:

I added a bit more to the prototype, including the ILI touch LCD screen. Additionally, I added more sensors. I did not like the breadboard prototype as connection became lost and the wires would disconnect very easily. So, I thought a solder prototype would make more sense, so I integrated everything on a proto board.

Prototype WORKED on 10/18/2026. Basic Functionality like lighting up, capacitive touch pad reducing brightness on LED, VEML dictating how bright LED should be.